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ACIDOGENIC POTENTIAL OF ORAL VIRIDANS STREPTOCOCCI IN RESPONSE TO GLUCOSE AT VARIOUS pH LEVELS AND DIFFERENT CONCENTRATIONS OF FLUORIDE

TANVEER ABBAS AND PERWEEN TARIQ
Department of Microbiology
University of Karachi, Karachi-75270, Pakistan

SUMMARY

One hundred and seventy five (175) isolates of viridans streptococci obtained from oral cavity of apparently healthy subjects were used in the present study to evaluate their potential to produce acid from glucose in the presence of various concentrations (0.1%, 0.05% and 0.01%) of sodium fluoride (NaF) at different pH levels (pH 5, 7 and 8). Nutrient broth containing 5% glucose at different pH levels were used to evaluate the acidogenic potential of these isolates in the absence of NaF.

It was observed that maximum acid production by overall number of isolates of viridans streptococci was at pH 7.0. It was 1.15 ml (average) in response to 5% glucose without NaF while the minimum acidity was produced at pH 5.0 (i.e. 0.77 ml average).

The overall results showed that the reduction in the titratable acidity was directly proportional to the concentration of NaF and acidogenic potential was reduced also found to be pH dependent and reduction was maximum at pH 5 with 0.1% and 0.05% of NaF. The reduction in acid production was lowest at pH 7.0.

INTRODUCTION

Among the members of the oral flora, the viridans streptococci are the most important causative agent of dental plaque, dental carries and periodontal disease (Duguid et al., 1978). These diseases are probably among the most common oral bacterial diseases in humans. However, their non-life threatening nature have minimized their importance in overall human health, but the economic burden for the treatment of these dental infections can be substantial (Loesche, 1986). Dental caries is still one of the major cause of loss of teeth. They key factors in dental caries are dental plaques containing acidogenic bacteria, fermentable carbohydrates and susceptible teeth (Cawson, 1991).

The viridans streptococci utilize fermentable carbohydrates and produce organic acids and extracellular polysaccharide (glucan). The acid and glucan production are important factors in the development of caries as this acid results in lowering of pH (below 5.0) which causes decalcification of tooth surface while extracellular polysaccharide (glucan) helps the bacteria in attachment to the tooth surface.

The production of acidity and subsequent cariogenicity of carbohydrate is dependent on the type of carbohydrates, its form, consistency, quantity and frequency of intake. The more refined carbohydrates such as glucose, fructose and sucrose, are more cariogenic than the crude form of carbohydrates such as starchs, because these are easily fermentable by bacteria to produce acid (Perera, 1984). Frequent consumption of fermentable carbohydrate keeps the acid level high for prolonged period and it is evident from bulk of research that acid production by oral flora is important, if not essential, for the development of dental diseases such as dental caries, periodontal disease etc. (Duguid et al., 1978; Cawson 1991; Jawetz et al., 1995).

Various measures have been proposed for the prevention of dental diseases. These include antibacterial measures; dietary measures (control of carbohydrates intake) modification in plaque metabolism and increasing resistance to the tooth attack (Cawson, 1991). One of the important measure to increase the resistance of the tooth to carious attack is the use of fluoride. Fluoride not only enhance the tooth resistance but also change the flora by elimination of potentially pathogenic microorganisms (Loesche et al., 1975; Abayaratna 1984; Cawson 1991). The exact mode of action of fluoride is not fully understood. It is believed that it protects teeth by promoting remineralization of enamel after the carious attack, by making enamel more resistant to dissolution by acid and by interfering with bacterial metabolism in the plaque (Russel and Melville, 1978).

As it is clear that the production of acid from the carbohydrate is the initial requirement for the cariogenic bacteria to initiate dental caries, the present study has been carried out to determine the effect of fluoride at different pH levels on acidogenic potential of viridans streptococci in response to glucose.

MATERIALS AND METHODS

Isolates:

One hundred and seventy five (175) isolates obtained from apparently healthy individuals were used to study the effects of NaF on their acidogenic potential in response to glucose at various pH levels.

Maintenance of culture:

All cultures were maintained on sodium azide blood agar medium (Sonnenwirth and Jarett, 1980).

Base Medium:

Nutrient broth (Oxoid) containing 5 gram of glucose per 100 ml of nutrient broth was used as base medium.

Concentrations of sodium fluoride (NaF)

NaF was incorporated into the base medium to get final media concentrations of 0.01%, 0.05% and 0.1%.

Control:

Nutrient broth containing no glucose, no NaF and nutrient broth containing different concentrations of NaF (0.01%, 0.05% and 0.1%) were used as controls.

Adjustment of pH:

The pH of base medium, controls and base medium containing different concentrations of sodium fluoride were adjusted with NaOH or HCl to get initial media pH measurements of 5.0, 7.0 and 8.0.

Dispensation of media:

Controls, base medium and base media containing different concentrations of sodium fluoride (NaF) at various pH levels were dispensed in 10 ml quantity in 18x150 mm tubes.

Preparation of inoculum:

Trypticase soy broth (Oxoid) was used for the preparation of inoculum. A loopful of culture was inoculated in 2.0 ml of trypticase soy broth and incubated at 35-37°C for 18-24 hours. Next day the inoculum was standardized by matching its turbidity with McFarland's Tube No.4 (Sonnenwirth and Jarett, 1980; Baron and Finegold, 1990).

Inoculation:

Control tubes and nutrient broth tubes containing 5% glucose and different concentration of sodium fluoride at different pH levels were inoculated with 0.1 ml of standardized inoculum.

Incubation:

All the inoculated tubes were incubated at 35-37°C for 48 hours.

Titration of acidity:

Bromocresol purple (1.6% was used as indicator. One drop of indicator was added to each tube. Development of yellow colour was indicative of acid production while development of purple colour was indicative of no acid production. 0.1N NaOH was used as base for the titration of acidity. 0.1N NaOH was added dropwise until a purplish tinge was obtained. Development of purplish tinge was indicative of neutralization of acid and was considered end point. Milliliters of 0.1N NaOH required for the neutralization of acid were recorded.

RESULTS AND DISCUSSION

Viridans streptococci are normal inhabitants of the oral respiratory and gastrointestinal mucosa. They have been considered harmless and of low virulence (Baron and Finegold, 1990). Under altered condition, they may act as opportunistic challengers and cause a number of diseases like dental plaques, dental caries, periodontal disease, dentoalveolar infections, perioral abscess, recurrent aphthous stomatitis (Drucker and Green 1978; Moore-Gillion et al., 1981; Williams et al., 1983; Gossling 1998; Baron and Finegold 1990; Hardie 1992).

The most common diseases caused by oral streptococci are dental caries and periodontal disease (Williams et al., 1983; Loesche 1986; Hardie 1992).

The development of caries by these organisms is dependent on two factors; firstly production of extracellular polysaccharide by utilization of fermentable carbohydrate and secondly, production of acid from the carbohydrates. The acid production causes the decalcification of tooth and initiates the process of tooth decay (Jenkinson 1989; Cawson 1991).

Table I
Pattern of acidogenic potential of overall number of viridans streptococci in response to various concentrations of NaF different pH levels

S.No.	Concentration of glucose and/or NaF	Average titratable acidity (Average ml of 0.1N NaOH) at pH levels		
		pH 5	pH 7	pH 8
1.	No Glucose + No NaF	0.00	0.00	0.00
2.	No Glucose + 0.1% NaF	0.00	0.00	0.00
3.	No Glucose + 0.05% NaF	0.00	0.00	0.00
4.	No Glucose + 0.01% NaF	0.00	0.00	0.00
5.	5% Glucose + No NaF	0.77	1.15	0.94
6.	5% Glucose + 0.1% NaF	0.23	0.41	0.31
7.	5% Glucose + 0.05% NaF	0.29	0.52	0.41
8.	5% Glucose + 0.01 NaF	0.37	0.62	0.52

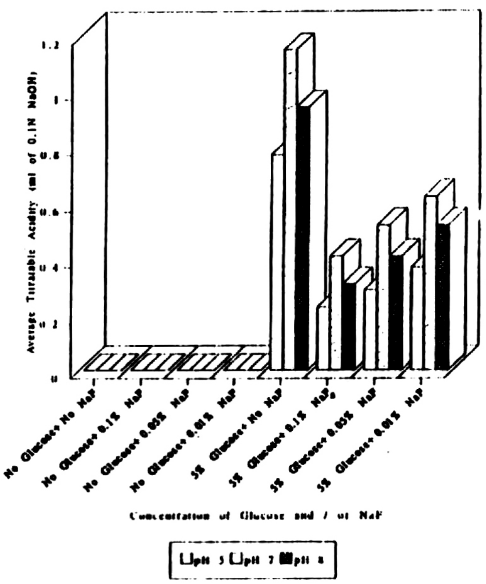

Fig.I: Pattern of acidogenic potential of overall number of *Viridans streptococci* in response to glucose with various concentrations of NaF at different pH levels.

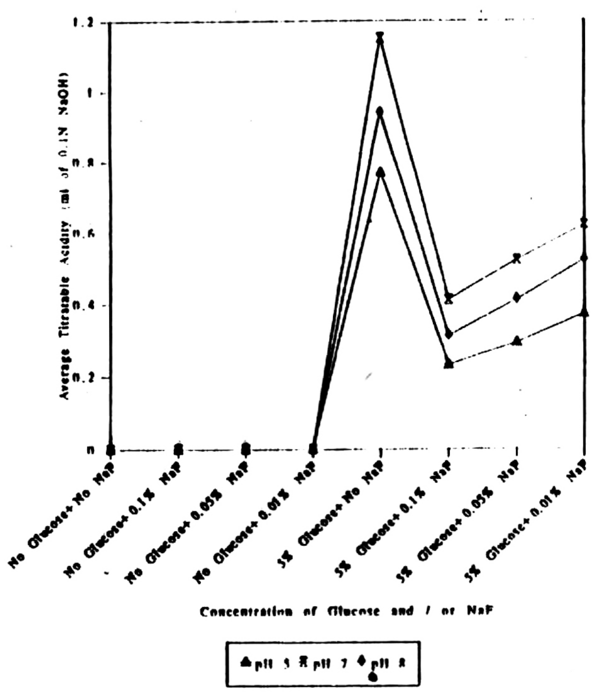

Fig.II: Pattern of acidogenic potential of overall number of *Viridans streptococci* in response to glucose with various concentration of NaF at different pH levels.

The most popular measure to control the dental caries to alter the microbial flora by elimination of the pathogenic microorganisms and by the enhancement of tooth resistance against the carious attack. In this regard, fluorides have been used because it can inhibit streptococci selectively(Loesche et al., 1975; Cawson 1991).

In present study; 175 isolates of oral viridans streptococci were used. Table I, Fig.I and II show the pattern of acidogenic potential of overall number of oral viridans streptococci in response to various concentrations of NaF at different pH levels. No titratable acid was observed when glucose was absent and only nutrient broth was utilized. It is obvious that acid was not produced because fermentable sugar required for acid production was lacking in the medium. For the same reason the acid production was not observed in the medium with various concentrations of NaF without glucose. All these served as negative control.

Overall number of viridans streptococci produced maximum titratable acidity at pH 7.0 (average 1.15 ml) which was followed by titratable acidity at pH 8i.e. 0.94 ml (average) and the minimum acidity was observed at pH i.e. 0.77 ml (average). The reason might be that production of acid lowered down the pH which resulted in loss of viability of bacteria and, in turn, reduction in acidogenic potential of these organisms.

However, when pH is lowered below 5, the risk of dissolution of enamel and superficial dentine increases and open the path for aciduric bacteria which causes more damage to dentine. The pH in plaque can drop below 5.0 for 30-120 minutes following exposure to a fermentable carbohydrate (Schachtele and Jensen, 1982). Most of the plaque forming bacteria are not metabolically active at this pH value (Denepitiya and Kleinberg, 1984) which might account for the minimum average acid production at pH 5.0 in this study. Growth and acid production of viridans streptococci have not been sufficiently studied at pH 8. Although Handleman and Kreinces (1973) studied the effect of pH on acid production and growth of *S. mutans* but they supplemented the media with phosphate. In present study the acidogenic potential of all isolates of viridans streptococci was reduced at pH 8.0 as compared to the pH 7.0. This might be due to the fact that pH 8.0 is not the optimum pH for growth and acid production and the growth is slow at pH 8.

The overall pattern of acidogenic potential in the presence of 5% glucose supplemented with various concentration of NaF showed that average acidogenic potential reduced with increasing concentration of NaF used at any pH level. Maximum acid was produced by pH 7.0 at all concentrations of NaF in comparison to average acid produced at the same concentration of NaF at the other pH levels. Maximum reduction+ in the acid production occurred at pH 5.0, however, the decrease in the concentration of the NaF at the same pH i.e. 5.0 resulted in the increase of acid production.

It is evident from literature that an acidic pH increases the antimicrobial effect of fluoride (Wright and Jenkins, 1954; Hamilton, 197; Marsh and Phillip, 1991). For example, low levels (1 millimole/litre) of NaF had little effect on microflora at neutral pH. However, when the pH fell down following carbohydrate metablolism, 1 millimole of NaF slowed acid production (Marsh and Phillip, 1991). In another study Hamilton and Ellwood (1978) observed that 160 ppm fluoride inhibited glycolysi at pH 6.5. When pH was returned to normal, the inhibition was reversed. It has been observed that if the extracellular environment become more acidic i.e. pH below 5, then the efficiency of the NaF, such as 5-10 ppm (5-10 microgram/ml) had inhibitory effect (Wright and Jenkins, 1954).

Sodium fluoride also affects production of polysaccharide by bacteria. In this connection, Shimura (1978) observed that production of water-insoluble fraction of polysaccharide by bacteria was inhibited in the presence of NaF in the growth medium, while the water-soluble fraction remained undisturbed. The adsorption of the bacterial polysaccharide on hydroxyapatite decreased with the decrease in production of water insoluble fraction. Thus NaF prevented the caries development by inhibiting production of water insoluble polysaccharide which is required for bacterial attachment.

It is clear from the present study that effectiveness of fluoride against the acidogenic potential of viridans streptococci is influenced by the concentration of NaF as well as the pH. Thus, beside concentration of fluoride, a pH change may also increase or decrease the effectiveness of fluoride. It is obvious that reduction in effectiveness of fluoride is beneficial for cariogenic bacteria as it has been well documented that acid production plays a crucial role in initiation of dental caries. Therefore, it will be feasible if, along with fluoride other preventive measures such as regular tooth brushing, mouthwashing, use of dental floss or toothpicks should also be practiced. Furthermore, between meal excessive and frequent intake of carbohydrates especially sugar should be avoided to provide a real contribution to dental health.

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